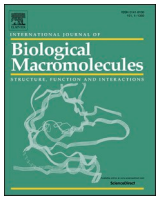
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Exploration of biomass fractionation and lignin removal for enhancing enzymatic digestion of wheat-stalk through deep eutectic solvent Cetyl trimethyl ammonium chloride:Lactic acid treatment

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ABSTRACT

The cationic surfactant-based deep eutectic solvents (DESs) have attracted extensive attention due to their effectual destruction of the natural anti-degradation barrier structure in lignocellulose. In this research, functional DES Cetyl trimethyl ammonium chloride:Lactic acid (CTAC:LA molar ratio 1:0.5 to 1:6) was fabricated for pretreating wheat-stalk. The relationships of accessibility, lignin removal, xylan separation, and enzymolysis efficiency were explored. The highest delignification (88.3 %) and xylan removal (89.0 %) were obtained through the treatment with CTAC:LA (1:4, mol/mol, 160 °C, 60 min), acquiring 86.8 % of enzymolysis efficiency. The structure of CTAC:LA-treated wheat-stalk was changed to porous state, while the accessibility and crystallinity were substantially improved to 668.2 mg/g and 57.6 %, respectively. The lignin surface area declined from 672.7 to 335.4 m^2/g. Furthermore, the structure of lignin disrupted by CTAC:LA was analyzed by 2D-HSQC NMR, implying that CTAC:LA could cleave the C—O covalent bond and C—C bond and degrade the S- and H-units in wheat-stalk lignin through interaction. The potential pretreatment mechanism was proposed through comprehensive exploration at the molecular level and macro level, and this built pretreatment process held great promise for valorizing biomass into highly valuable chemicals.

1. Introduction

Under the double pressure of global energy crisis and environmental pollution issue, seeking effectual and eco-friendly energy production mode has become a research hotspot in modern society [1]. Wheat-stalk (WS) is a common agricultural waste, rendering as an important feedstock for producing chemicals and fuels in biorefinery [2,3]. However, in the complex structure of WS, lignin combines with hemicelluloses to fabricate the lignin-carbohydrate complex (LCC), shielding cellulose accessible surface, blocking the cell wall deconstruction and hindering enzymolysis [4]. Distinct from cellulose and lignin, hemicellulose is prone to degradation under high temperature [5]. Lignin is known as a polyphenol biomacromolecule with complex structure [6]. It has a random three-dimensional structure formed by covalent cross-linking of multiple subunits, and the complexity of these connections makes lignin difficult to degrade [7,8]. Consequently, it is of great significance to disrupt lignin and hemicellulose in lignocellulose through efficient pretreatment, greatly improving its enzymolysis efficiency.

In recent years, deep eutectic solvents (DESs), which can be fabricated by mixing hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD), have gained extensive attentions as a new green solvent for breaking down and hydrolyzing LCC through the disruption of the existing bonding interactions between carbohydrates and lignin [9,10]. Alvarez-Vasco et al. used four DESs (ChCl:Lactic acid, ChCl:Acetic acid, ChCl:Levulinic acid, and ChCl:Glycerol) for extraction of lignin in poplar wood. Among all four DESs, ChCl:Lactic acid had the greatest effect on lignin removal [11]. It has been reported that the enhanced lignin removal effect of ChCl:Lactic acid could be attributed to the high hydrogen bond supply capacity of lactic acid (LA), which makes the ether bonds in lignin easy to crack during the removal of lignin [12]. Recently, cationic surfactants with hydrophobic and hydrophilic properties have been used to reduce surface tension of lignocellulose driving the removal of hydrophobic molecules such as lignin, thus affecting the enzymolysis efficiency of lignocellulose [13], and they were used to synthesize DESs for removing lignin and enhancing enzymatic saccharification. DES synthesized with cetyl trimethyl ammonium

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bromide (CTAB) and lactic acid (LA) was utilized to effectually eliminate lignin (51.5 %) and xylan (79.9 %) from rice husk, and the enzymolysis efficiency of treated rice husk reached 5 times that of the raw material [14]. Using CTAB and ethylene glycol to synthesize DES for the pretreatment of rape straw, the lignin removal reached 62.2 %, resulting in a 46.0 % of enzymolysis efficiency [15]. However, the pretreatment ability of CTAB-based DES containing Br^- is not ideal, and exploring high efficiency of surfactant-based DES for efficiently removing lignin and promoting enzymolysis has gained great attention. In addition, the corresponding mechanism for lignin elimination and enhanced enzymolysis through the DES pretreatment need to be well interpreted.

It has been reported that the supplementary of Cl^- into the pretreatment system can significantly increase the lignin elimination [9,16] and destroy lignocellulose structure [17]. In addition, chloride can participate in stabilizing the protonation transition state of the acid-catalyzed reaction [18]. Therefore, cationic surfactant-based DESs containing Cl^- group might be efficiently used for efficient biomass pretreatment and valorization. Accordingly, cetyl trimethyl ammonium chloride (CTAC), as Cl^--contained surfactant, was utilized as HBA to fabricate novel surfactant-based DES CTAC:LA by using lactic acid (LA) as HBD, which was used as an eco-friendly and efficient solvent for WS pretreatment in this research. The impact of pretreatment parameters such as DES CTAC:LA solvent properties, CTAC-to-LA molar ratio, pretreatment temperature, and pretreatment duration on biomass fractionation efficiency were assessed. Meanwhile, the contents of furfural and XOSs and the effects of reaction severity factors on their formation were analyzed. The alterations in the chemical composition, surface microstructure, crystal structure, accessibility, and surface lignin area of WS through the CTAC:LA treatment were interpreted by a range of characterization methods (XRD, SEM, dye adsorption techniques, etc.). Additionally, two-dimensional heteronuclear single quantum coherent nuclear magnetic resonance (2D-HSQC NMR) was applied to analyze the changes of lignin structure. The CTAC:LA treatment mechanism for enhancing the enzymolysis of WS was explored and discussed. These efforts comprehensively elaborated the CTAC:LA treatment mechanism, deepened the understanding of the structural characteristics of lignin, and provided scientific basis for promoting lignocellulose pretreatment efficiency in biorefining.

2. Materials and methods

2.1. Chemicals and materials

The wheat-stalk (WS) used in the experiment was collected from Jiangsu, China. The substrates were ground to 20–60 mesh. Cetyl trimethyl ammonium chloride (CTAC) were purchased from Aladdin Biochemical Technology Co. Ltd. Lactic acid (AR) and other chemicals were sourced from Sinopharm Chemical Reagent Co., LTD.

2.2. Preparation of CTAC:LA and related parameters

DES was fabricated by cetyl trimethyl ammonium chloride (HBA) and lactic acid (HBD) at different HBA:HBD molar ratios (from 1:0.5 to 1:6), the surface tension was measured with surface tensimeter (KINO A201, U.S.A.), the viscosity was measured with rotary viscometer (Brookfield A1017, U.S.A.), and the average was repeated three times. The synthesis of DES (CTAC:LA) was verified by FT-IR. The hydrogen bond capacity of DESs was quantified by Kamlet-Taft solvation parameters, and hydrogen bond donor capacity (α), hydrogen bond acceptor capacity (β), and polarity/polarizability (π*) were determined as previously described [19].

2.3. Pretreatment and chemical composition analysis of WS

DES and WS were mixed at a specific solid-liquid mass ratio (1:15) in a 250-mL flask, which was immersed in oil bath. The pretreatment was implemented with DES (molar ratio of HBA-to-HBD 1:0.5–1:6) at a designed reaction temperature (120 °C–200 °C) for a certain pretreatment duration (20 min–100 min). The pretreatment severity factor ($LogR_0$) serves as a comprehensive parameter for assessing the intensity of biomass pretreatment processes, and this factor typically considers the effects of temperature and time during pretreatment on reaction strength. It has proven to be an effective tool for comparing and predicting the performance of various pretreatment strategies in terms of hemicellulose dissolution and lignin removal [20]. Based on the given reaction temperature (T) and time (t), $LogR_0$ (the severity factor) was defined according to Eq. (1). The pretreated WS samples obtained from the different severity factor were used for chemical composition analysis and enzymolysis.

$$LogR_0 = Log\left[t \times exp\left(\frac{T-100}{14.75}\right)\right] \quad (1)$$

According to the analytical method provided by the National Renewable Energy Laboratory of the United States, the biomass chemical composition (glucan, xylan, and lignin) was determined. The degree of the recovery of solid (R_S), removal of xylan (R_X), and delignification (D) were calculated as below:

$$R_S\ (\%) = \frac{Pretreated\ WS\ (g)}{Unpretreated\ WS\ (g)} \times 100 \quad (2)$$

$$D\ (\%) = \left(1 - \frac{Lignin\ in\ pretreated\ WS\ (g)}{Lignin\ in\ unpretreated\ WS\ (g)}\right) \times 100 \quad (3)$$

$$R_X\ (\%) = \left(1 - \frac{Xylan\ in\ pretreated\ WS\ (g)}{Xylan\ in\ unpretreated\ WS\ (g)}\right) \times 100 \quad (4)$$

2.4. Enzymolysis of WS

Raw WS or WS after pretreatment were hydrolyzed with cellulase (10 FPU/g-glucan) in citric acid buffer solution (pH 4.8, 50 mM) to test enzymolysis efficiency. The hydrolysis reaction was implemented at 50 °C for 72 h. The supernatant was subsequently sampled to determine the monosaccharide content. Enzymolysis efficiency (EE) was calculated using the following formula:

$$EE\ (\%) = \frac{[Glucose] + [cellobiose]\ in\ enzymatic\ hydrolysate\ (g) \times 0.9}{Glucan\ in\ pretreated\ WS\ (g)} \times 100 \quad (5)$$

2.5. Quantum chemical calculations and molecular dynamics (MD) simulations

To gain a deeper understanding of the interactions between the typical structural units [S- (syringyl), G- (guaiacyl), and H- (*p*-hydroxyphenyl) types] of WS lignin and CTAC:LA solvent systems, the interaction types were visually differentiated based on the independent gradient model (IGM) in density functional theory (DFT) to reveal the strong non-covalent interactions between atoms or molecules in the system [21]. Additionally, visual molecular dynamics (VMD) was utilized to draw electrostatic potential (ESP) and other color mapping isosurfaces to visualize the trajectory data of the system [22].

2.6. Separation and characterization analysis of lignin

The extraction method of lignin was referred to the method as previously reported [23]. The separated pure lignin fractions were characterized by two-dimensional heteronuclear single quantum coherence nuclear magnetic resonance (2D-HSQC NMR) (Bruker II 600 MHz Spectrometer, Germany). For quantitative analysis methods, selected "aromatic units" as "I". The choice of the $0.5IS_{2,6} + IG_2 + 0.5IH_{2,6}$ signals as I for WS lignin.

$$IC_9 \text{ units} = 0.5 \times IS_{2,6} + IG_2 + 0.5 \times IH_{2,6} \quad (6)$$

where $IS_{2,6}$ is the integration of $S_{2,6}$ and $S'_{2,6}$, IG_2 is the integral value of G_2. $IH_{2,6}$ is the integration of $H_{2,6}$. IC_9 represents the integral value of the aromatic ring (S + H + G). According to the internal standard (IC_9), the amount of IX% could be obtained by the following formula [24].

$$IX\ (\%) = \frac{IX}{IC_9} \times 100 \quad (7)$$

where IX is the position integral value of $S_{2,6}$, G, $H_{2,6}$, A (*β-O-4*), B (*β-β*), C (*β-5*), etc. in particular. S/G ratio was calculated according to the following formula [25].

$$S \Big/ G = \frac{0.5 \times IS_{2,6}}{IG} \quad (8)$$

where IS (the quantity of S unit) is half of $IS_{2,6}$ and IG is the quantity of G unit. For the convenience of comparison, the molecule (quantity of IS unit) is normalized.

2.7. Properties of WS after pretreatment

2.7.1. Accessibility and lignin surface area

Lignocellulose accessibility was measured using Congo red dye adsorption. The surface area of raw and treated WS lignin was investigated through the adsorption of Azure B dyes at different concentrations [26]. Langmuir nonlinear fitting was used to assess the maximum dye adsorption capacity of biomass samples and evaluate its lignocellulose accessibility and lignin surface area [27].

2.7.2. FT-IR, XRD and SEM

FT-IR was applied to observe the structural difference of the compound molecules in the range of 4000–500 cm^{-1}, and the analysis was performed on instrument (Thermo Fisher Nicolet IS50) using the KBr pressure tablet method. The structure and morphology of the WS samples were characterized by scanning electron microscopy (SEM) (JSM-6360LA, Japan), the surface of the WS sample was sprayed with gold and imaged at 5 kV. The crystallinity was measured by X-ray diffractometer (XRD) (SmartLab SE, Rigaku, Japan) with diffraction angles ranging from 5° to 80°. The crystallinity index (*CrI*) was calculated [28].

2.8. Recovery and reuse of the DES

The recycling ability of DES is an important index of its biomass pretreatment performance [14]. Firstly, WS was pretreated with CTAC:LA (HBA:HBD molar ratio = 1:4) at 160 °C at a 1:15 of solid-liquid mass ratio for 60 min. After the reaction, the pretreated liquid was filtered using filter bottle and Brinell funnel. Then the filtrate was repeatedly used as a solvent for the next run of pretreatment, which was carried out under the same biomass conditions. The filter residue was rinsed and utilized for subsequent component analysis and enzymolysis.

2.9. Statistical analysis

All the pretreatment and enzymolysis experiments were implemented in triplicate to confirm the reproducibility of the data. The SPSS software was utilized for the analysis of variance (ANOVA) to test for statistically significant differences.

3. Results and discussion

3.1. Investigation of HBA:HBD molar ratio on affecting the enzymolysis

The molar ratio of HBA to HBD in DES can affect the solubility and selective dissolution of biomass components [29]. It was found that the total lignin content (acid-soluble lignin + acid-insoluble lignin) reached 14.6 % (Fig. 1a), the lignin removal only reached 51.4 %, and the enzymolysis efficiency was 49.4 % after CTAC:LA (HBA:HBD = 1:0.5, 160 °C, 60 min) treatment (Fig. 1b). It was reported that a low proportion of HBD couldn't form effective hydrogen-bond network and resulted in increased viscosity and reduced fluidity of the solvent, which would impact the dissolution efficiency between the solvent and biomass [30]. When the CTAC-to-LA molar ratio reached 1:4, the xylan removal (89.0 %) and lignin removal (88.3 %) reached their highest value, acquiring 86.8 % of enzymolysis efficiency. As the CTAC-to-LA molar ratio continued to increase to 1:6, the lignin removal showed a weak downward trend (from 88.3 % to 86.5 %), and the enzymolysis efficiency also declined correspondingly (Fig. 1b). Simultaneously, it has been demonstrated that the molar ratio of HBA to HBD significantly impacted the enzymolysis efficiency through statistical data analysis ($p < 0.05$) (Table S1, see supplementary file). Continuing to increase the molar ratio of HBA:HBD might improve the polarity of DESs and weaken its interaction with non-polar or hydrophobic biomass components, such as lignin, thereby reducing DES's ability to dissolve lignocellulose [31]. Therefore, CTAC:LA = 1:4 was considered the most appropriate HBA:HBD molar ratio.

By using LA as HBD, the absorption peak of LA's carboxyl group was red-shifted, as shown in FT-IR spectra. Meanwhile, the bending vibration of quaternary ammonium cation in CTAC also changed (Fig. S1, see supplementary file). The frequency shift and intensity changes of the representative bands of CTAC and LA in FT-IR spectra confirmed the existence of molecular hydrogen bonds between CTAC and LA [32], implying the successful synthesis of DES. The Kamlet-Taft solvation parameters of DES solvent can be used to evaluate its solubility [33]. With increasing the molar ratio of HBA:HBD, the π* value remained between 0.20 and 0.27, the β value of the solvent (Lewis alkalinity) remained between 1.13 and 1.20, manifesting great hydrogen bond acceptance capacity of CTAC. The α value of the solvent (Lewis acidity) elevated from 0.36 to 1.77, and the hydrogen bond donor capacity changed with the HBA:HBD molar ratio (from 1:0.5 to 1:6) (Table S2, see supplementary file). It was also found that there were fittings about the α value and the solubility of lignin and xylan. As displayed in Fig. 1c, the xylan removal [$Y_{(\text{Xylan removal})} = 46.93 \times X_{(\text{Hydrogen bond acidity})} + 31.12$, $R^2 = 0.86$] and delignification [$Y_{(\text{Delignification})} = 39.34 \times X_{(\text{Hydrogen bond acidity})} + 39.76$, $R^2 = 0.91$] were positively correlated with the α value of DES. Higher α value in the solvent facilitated the elimination of lignin and xylan because it might enhance the hydrogen-bond interaction between the solvent and lignocellulose, making it easier for the lignin and xylan to dissociate from the biomass complex [34].

The viscosity and surface tension of the solvent can directly affect its diffusion rate and permeability in lignocellulosic pores [35]. CTAC is a cationic surfactant with molecular structure that includes hydrophilic head and hydrophobic tail, giving it the ability to reduce surface tension [36]. As showcased in Table 1, solvent systems with different CTAC:LA molar ratios all exhibited low surface tension (29.1–30.9 mN/m), which facilitated CTAC:LA in effectually wetting solid surfaces, penetrating biomass fibers more easily, and promoting the separation of lignocellulosic components. When the CTAC:LA molar ratio reached 1:2, the high viscosity (1223.1 cP) limited the effective contact between the solvent and the biomass. As the molar ratio of CTAC:LA was increased to 1:4, the viscosity was reduced to 380.9 cP, and the removal of lignin and xylan elevated apparently. This was because DESs with lower viscosity could penetrate into the biomass structure more quickly, improving mass transfer performance and efficiency [37].

3.2. Investigation of CTAC:LA treatment under the different severity factor

The severity factor of pretreatment plays decisive role in the overall effectiveness of the pretreatment [38]. Accordingly, the impact of two key parameters in the severity factor (pretreatment temperature and duration) on the pretreatment effectiveness was assessed and optimized.

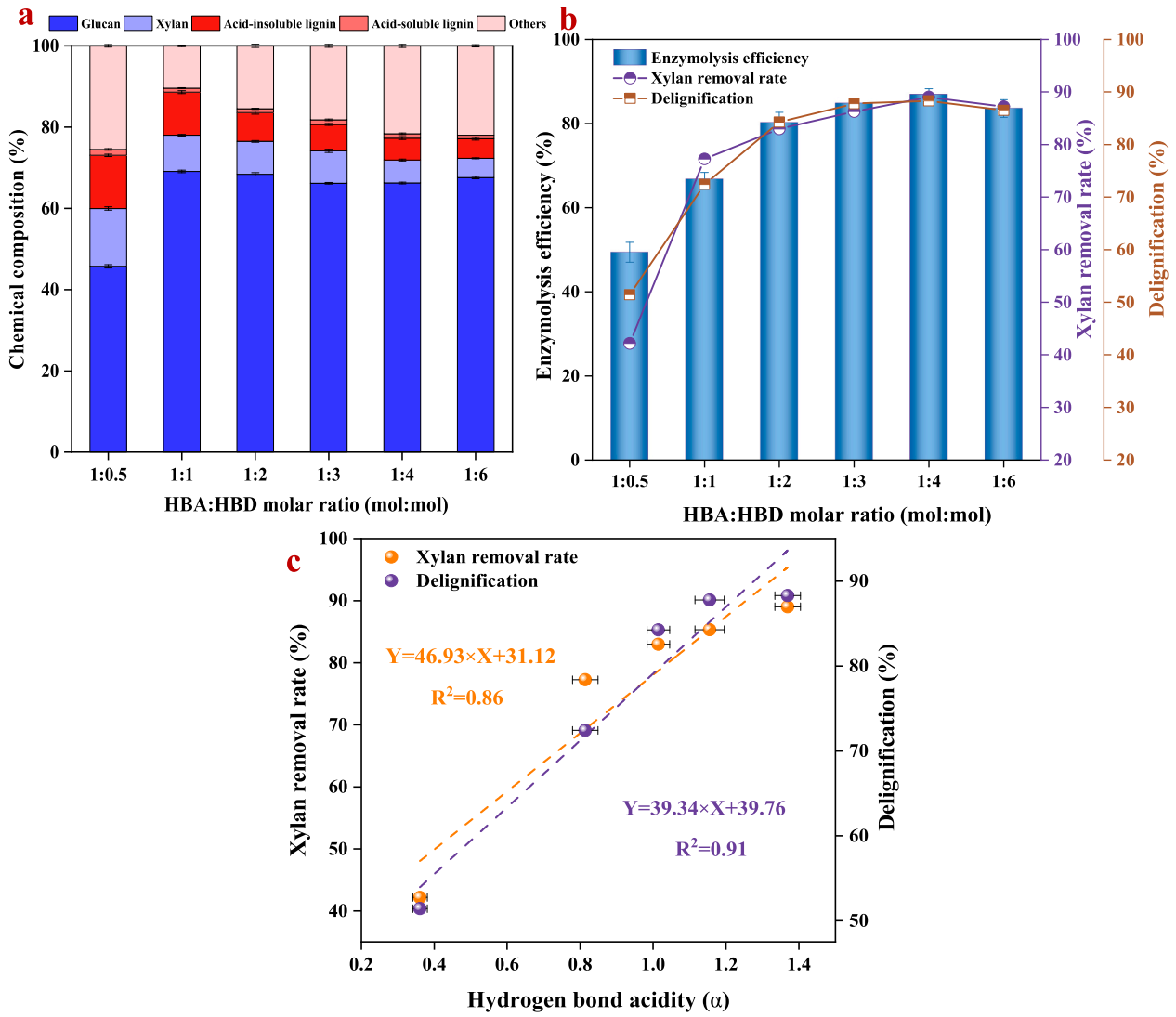


Fig. 1. Assessing the HBA:HBD molar ratio on affecting the chemical composition (a) and the enzymolysis efficiency of WS (b); The fitting of hydrogen-bond acidity (α) with delignification and xylan removal (c). [pretreatment conditions: 160 °C, 60 min].

Table 1
Viscosity and surface tension of CTAC:LA with different molar ratios.

HBA:HBD molar ratio	Viscosity, cP	Surface tension, mN/m
1:0.5	/	/
1:1	/	/
1:2	1223.1	29.1
1:3	641.7	30.3
1:4	380.9	30.1
1:6	212.0	30.9

As shown in Tables S3 & S4 (see supplementary files), the statistical data analysis implied that both pretreatment temperature and reaction duration significantly influenced enzymolysis efficiency ($p < 0.05$). Upon undergoing pretreatment temperature from 120 °C to 160 °C, there was significant decrease in lignin content (acid-soluble lignin + acid-insoluble lignin) in WS from 18.5 % to 8.0 %, and a decrease in xylan content from 15.1 % to 8.1 % (Fig. 2a). This resulted in substantial increase in the lignin removal (from 33.4 % to 84.3 %) and xylan removal (from 41.7 % to 83.0 %), while enzymolysis efficiency elevated from 49.4 % to 80.1 % (Fig. 2b). According to the Fig. 2b, the delignification increased from 84.3 % to 91.2 %, but the trend of improvement in enzymolysis efficiency was no longer obvious as the pretreatment temperature increased from 160 °C to 180 °C. Based on the above results, 160 °C was considered to be the most appropriate reaction temperature.

According to Fig. 2c & d, upon raising reaction duration from 20 min to 60 min, the lignin content (acid-soluble lignin + acid-insoluble lignin) decreased from 15.6 % to 8.0 %, the lignin removal elevated from 37.6 % to 84.3 %, the xylan removal rose from 50.5 % to 83.0 %, and the enzymolysis efficiency was raised from 55.8 % to 80.1 %. However, with the continuous increase of reaction duration (from 60 min to 100 min) and pretreatment temperature (from 180 °C to 200 °C), the enzymolysis efficiency still manifested downward trend, although the xylan separation and lignin removal were slightly increased. This phenomenon was associated with the fact that higher temperatures and prolonged pretreatment duration might accelerate biomass degradation and produce more by-products, and excessive by-products will bind to the surface of the material with other degradations, inhibiting the digestion of lignocellulose with cellulases [39]. Consequently, severity factor of 3.54 (160 °C, 60 min) was deemed the optimal factor for the CTAC:LA treatment of WS.

Pretreatment with certain severity factor might have a crucial influence on lignin removal and xylan separation [40]. Fig. 3b manifested that there was linear fitting about the delignification [$Y_{(Delignification)} = 26.51 \times X_{(Log\ R0)}$-22.68, $R^2 = 0.80$] and xylan removal [$Y_{(Xylan\ removal)} =$

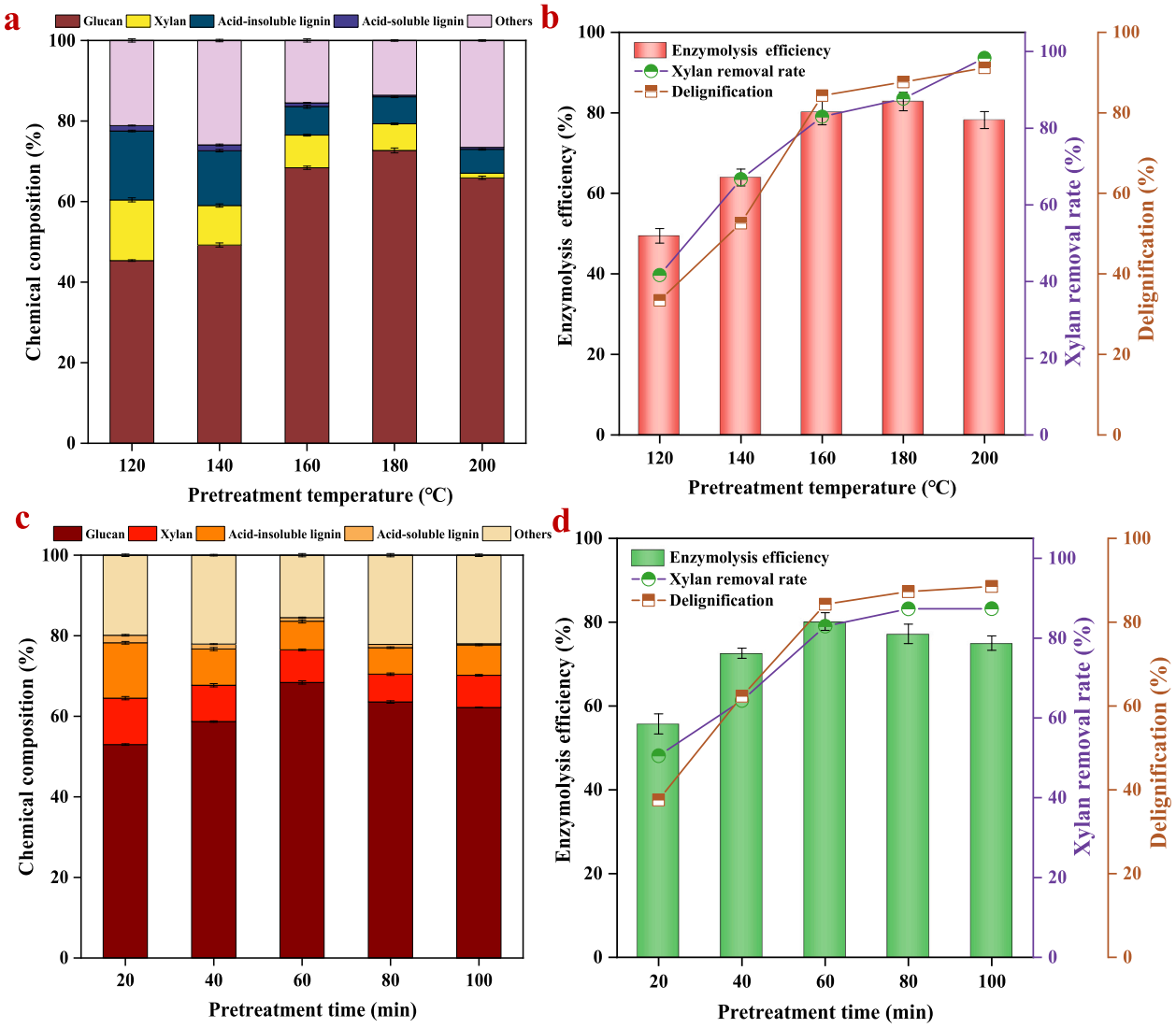


Fig. 2. Evaluating the pretreatment temperature on influencing the chemical composition (a) and enzymolysis efficiency of WS (b) [pretreatment conditions: 60 min, CTAC:LA = 1:2]; Assessing the pretreatment time on influencing the chemical composition (c) and enzymolysis efficiency of WS (d) [pretreatment conditions: 160 °C, CTAC:LA = 1:2].

23.58 × $X_{(Log\ R0)}$ -8.13, R^2 = 0.87] with the severity factor (Log R_0). With increasing pretreatment severity factor (Log R_0), both lignin and xylan separation also increased. These results manifested that the intensified pretreatment severity might contribute to the destruction of lignin and xylan, thereby enhancing biomass pretreatment efficiency and promoting the enzymolysis efficiency [41]. According to Fig. 3a, the efficiency of enzymolysis increased positively with the delignification [$Y_{(Enzymolysis\ efficiency)} = 0.49 \times X_{(Delignification)} + 36.00$, $R^2 = 0.93$] and xylan removal [$Y_{(Enzymolysis\ efficiency)} = 0.60 \times X_{(Xylan\ removal)} + 25.86$, $R^2 = 0.88$].

Xylo-oligosaccharides (XOSs) and furfural are valuable products that can be derived from biomass in the pretreatment process [42,43]. In this research, the relationship between the content of XOSs and furfural in the pretreatment solution with the severity factor by adjustment of pretreatment temperature and duration was investigated. As the severity factor elevated from 2.36 to 3.54, the xylan degradation led to an increase in XOSs production from 0.6 g/L to 2.3 g/L. However, excessively high temperatures and longer reaction duration might lead to further degradation of XOSs into xylose, demonstrating an inverse "hook-shape" fitting [$Y_{(Xylo\text{-}oligosaccharides)} = -0.60 \times X^3 + 5.72 \times X^2 - 16.36 \times X_{(Log\ R0)} + 15.23$, $R^2 = 0.82$] between the content of XOSs and severity factors (Fig. 3c). Furfural, as an important and versatile chemical intermediate, is acquired through the dehydration of xylose in acidic environment at high temperatures [44]. It could be seen from Fig. 3c that there was positive fitting between the severity factors and furfural production [$Y_{(Furfural)} = 3.21 \times X_{(Log\ R0)}$ -8.56, $R^2 = 0.84$]. With increasing severity factors from 2.96 to 4.72, the formed furfural could increase from 0.5 g/L to 6.0 g/L.

3.3. Analysis of lignin fractions after pretreatment by 2D-HSQC NMR spectra

2D-HSQC NMR was utilized to investigate the structural changes of lignin following DES (CTAC:LA) treatment to explore the mechanism of CTAC:LA on lignin. The lignin in raw material, CTAC-treated WS and LA-treated WS were employed to compare the structural alterations of lignin in CTAC:LA-treated WS (CTAC-to-LA molar ratio 1:4) [160 °C, 60 min]. The NMR spectra could be categorized into side chain (δ_C/δ_H 50–90/2.5–5.5) and aromatic (δ_C/δ_H 90–140/5.5–8.0) regions, with corresponding signal peaks listed in Table S5 (see supplementary files). Ether bond (β-O-4, A) emerged as the predominant substructure within lignin macromolecules, followed by phenylcoumaran (β-5, C), resinol

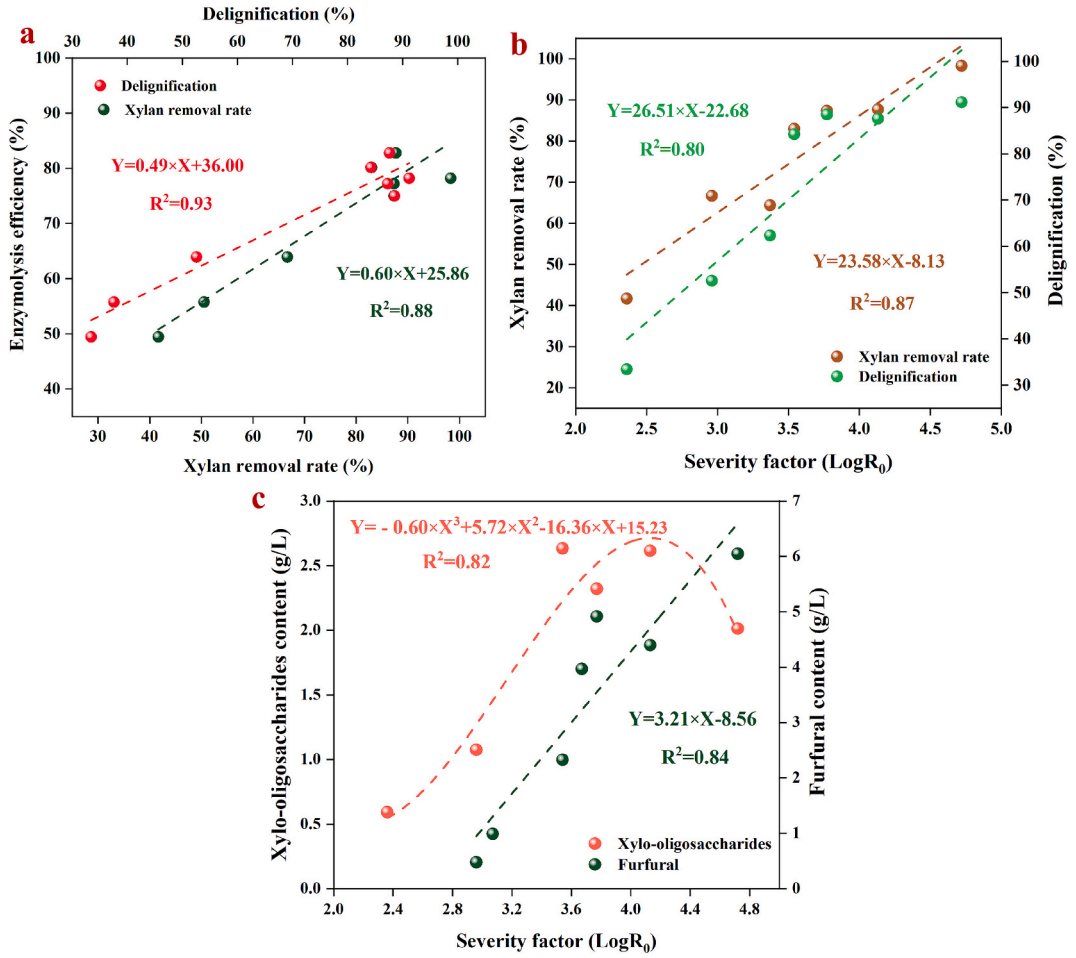


Fig. 3. The linear fitting about xylan removal and delignification with enzymolysis efficiency (a); The fitting of the severity factor ($LogR_0$) with delignification and xylan removal (b); Linear fitting about severity factor ($LogR_0$) with the contents of furfural and xylo-oligosaccharides (c).

(*β-β*, B), and other secondary bonds.

In the smooth side chain region (Fig. 4a, c, e & g), all lignin samples exhibited prominent methoxy (-OCH_3) peaks. The typical signal peak A_α (δ_C/δ_H 71.8/4.86) could be clearly seen in the lignin spectrum of the raw material, but the signal peak significantly weakened after CTAC and lactic acid (LA) pretreatment alone, and even disappeared after DES (CTAC:LA) pretreatment. The A_γ-site content (δ_C/δ_H 59.5/3.21–3.72) of the raw material was 17.9/100Ar, and signal attenuation was evident at the A_γ-site of the lignin. Following various pretreatment methods, there was significant change in the A_γ-site content to 17.6/100Ar (CTAC), 22.9/100Ar (LA), and 5.8/100Ar (CTAC:LA), respectively. It was also observed that the decomposition rate of *β*-O-4 connected to S units [A_β(S)] was higher than that of *β*-O-4 connected to G units [A_β(G)]. The cleavage of *β*-O-4 bond (A) was the main reaction to degrade lignin, the substantial removal of lignin was more likely to be attributed to the breaking of carbohydrate ester bonds during LA-mediated DES (CTAC:LA) treatment.

Additionally, apart from the cleavage of *β*-O-4, other carbon-carbon (C—C) bonds were also cleaved to a certain extent. The characteristic peak of *β*-5 disappeared after CTAC:LA treatment, and the peak strength of resinol (*β-β*, B) subjunction side chain bond also weakened. The original WS contained 5.7/100Ar of content B_β, which decreased to 1.2/100Ar after treatment with CTAC alone. However, after LA-mediated treatment, the signal peak was enhanced and the content increased to 3.3/100Ar, implying that partial lignin condensation reaction was involved in the LA treatment process [45]. The content dropped to 2.3/100Ar after CTAC:LA treatment, and no serious condensation reaction occurred. This explained the reason that CTAC as hydrogen bond acceptor for DES could achieve more efficient pretreatment efficiency.

In the aromatic regions, the syringyl (S) unit presented clearly correlated signal for the $S_{2,6}$ (δ_C/δ_H 104.1/6.74) (Fig. 4b, d, f & h). The correlated signals at δ_C/δ_H 111.7/6.95 (G_2), 115.8/6.76 (G_5), and 118.9/6.75 (G_6) were indicative of the presence of guaiacyl (G) units, the associated signal of *p*-hydroxyphenyl (H) (δ_C/δ_H 127.5/7.19) was also detected. The percentage of these structural units was determined by comparing their signals to the total number of aromatic rings (S + G + H) [46]. The signals of structural units (S, H) in lignin exhibited diminishing trend, while the proportion of G-structural units increased after employing various pretreatment strategies, and the proportions of structural units were detailed in Table 2. This was attributed to the acidic DES (CTAC:LA) treatment, which caused the S-unit to be more readily degraded than the G-unit. The greater stability of G-units resulted in an increased proportion of G-units in residual lignin [47]. After treatment with CTAC:LA, the aromatic groups of lignin were greatly eliminated, and only small groups could be observed after significant magnification, this highlighted the high delignification ability of DES solvent. Furthermore, the alteration of the S/G ratio in lignin served as a crucial factor in observing the structural alterations that occurred during DES treatment. It has been reported that a high S/G ratio of lignin might exert negative influence on the digestibility of lignocellulose [48]. Wang et al. discovered that phenyl-carbon ions, generated through the dehydration reaction of lignin under acidic conditions, readily condensed to form new C—C bonds, thereby increasing the S/G value [49]. The S/G ratio of raw lignin was remarkably high at 1:1.03, the S/G ratio of lignin reached 1:1.31 after LA pretreatment alone, but decreased to 1:2.45 after DES treatment, indicating that there

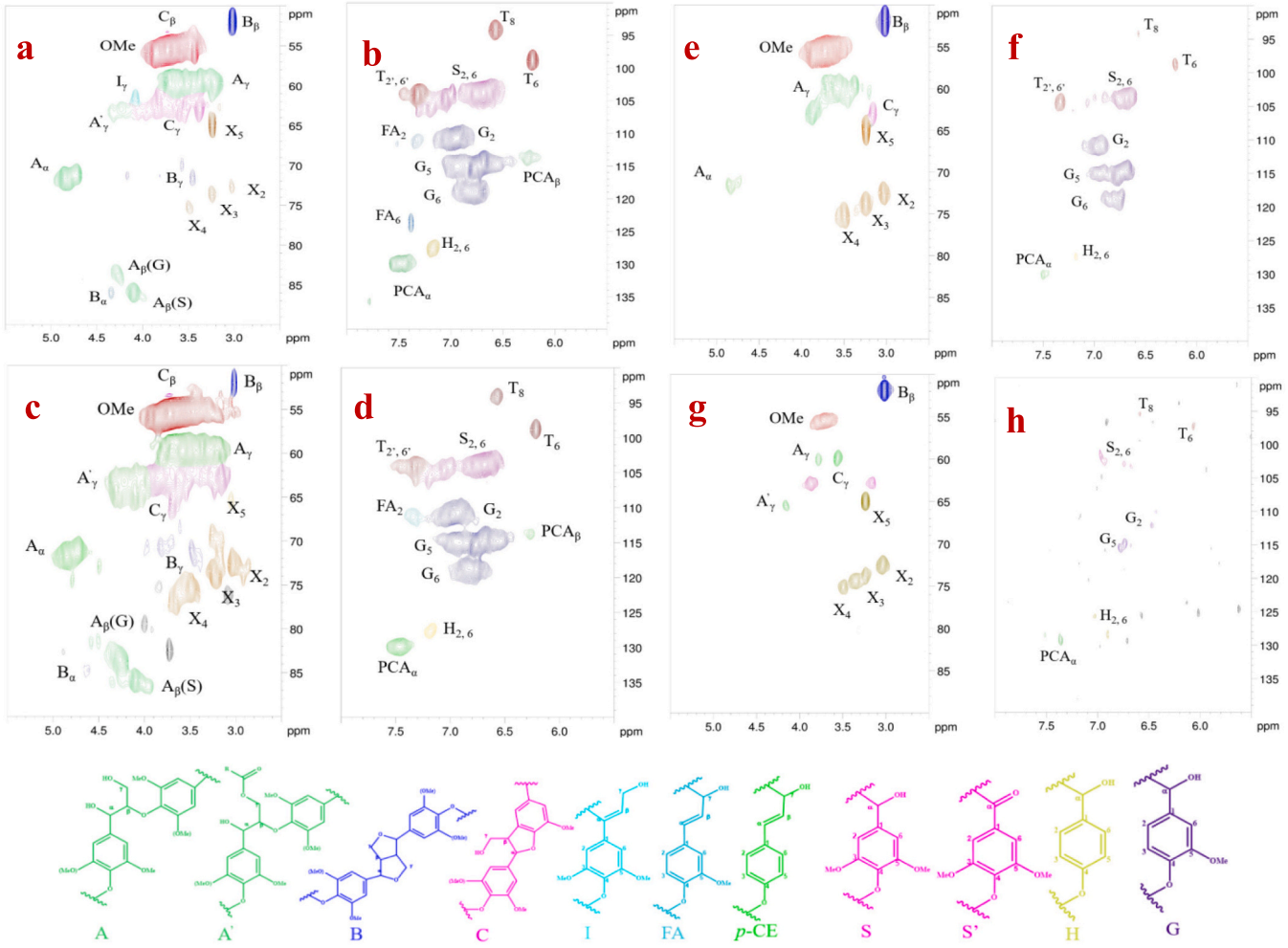


Fig. 4. 2D-HSQC NMR spectra of untreated WS-lignin (a, b) WS-lignin after CTAC pretreatment (c, d), WS-lignin after LA pretreatment (e, f), and WS-lignin after CTAC:LA pretreatment (g, h) [pretreatment condition: 160 ℃, 60 min].

Table 2
Content of lignin structural units in WS before and after pretreatment.

Samples	S (%)	G (%)	H (%)	S/G ratio
Raw WS	46.6	47.9	5.5	1:1.03
CTAC-treated WS	39.9	55.4	4.6	1:1.39
LA-treated WS	43.3	56.7	0.0	1:1.31
CTAC:LA-treated WS	29.0	71.0	0.0	1:2.45

was no serious condensation reaction of WS lignin after CTAC:LA treatment. Under the treatment of DES synthesized by LA and CTAC, a high lignin removal and effectual enzymolysis could be acquired by cleaving C—O and C—C bonds without causing significant lignin condensation reactions. The smaller lignin fragments produced by CTAC:LA treatment could be further processed into valuable chemicals such as phenols, cresol, and other aromatic compounds.

3.4. Molecular dynamics simulation

Molecular dynamics simulation is able to render atomic and molecular level insights by simulating the interactions between substrate and solvent molecules [50]. In this research, the interactions between the CTAC:LA treatment system and lignin were discussed by molecular dynamics simulation. The lignin in WS is a typical S-G-H type lignin (Syringyl-Guaiacyl-*p*-Hydroxyphenyl lignin). To further understand the interaction between the lignin unit and the CTAC:LA treatment system, the weak interaction between the CTAC:LA and the lignin structure was visualized. Van der Waals forces, which mainly include electrostatic interaction, induction force and dispersion force, might promote the interaction between pretreatment solvents and non-polar or hydrophobic compounds in lignocellulose, and promote their degradation [51]. In molecular simulations, electrostatic interactions can be evaluated by electrostatic potential. Among the three lignin structural models, G-type lignin model molecules manifested the strongest electrostatic interaction with DES, since the lowest self-electrostatic potential value of −0.961 (Fig. 5a). The hydroxy hydrogen atoms of lignin have a positive electrostatic potential, while the hydroxy oxygen atoms have the opposite. The natural affinity between the components of lignocellulose might partly be attributed to forming hydrogen-bond network between hydroxyl hydrogen and hydroxyl oxygen [52]. However, Cl^- in CTAC might introduce negative electrostatic potential, which would help CTAC:LA treatment system to disturb the hydrogen-bond network between lignocellulosic components and promote the stripping of lignin from the lignocellulose surface.

The Independent Gradient Model (IGM) can be utilized to elucidate robust non-covalent interactions between atoms or molecules within extensive systems [53]. Fig. 5b & c depicted the iso-surface images and scatter diagrams of IGM respectively. The iso-surface images clearly demonstrated that H-type lignin structure exhibited the strongest hydrogen-bond interaction with the pretreatment system (depicted by

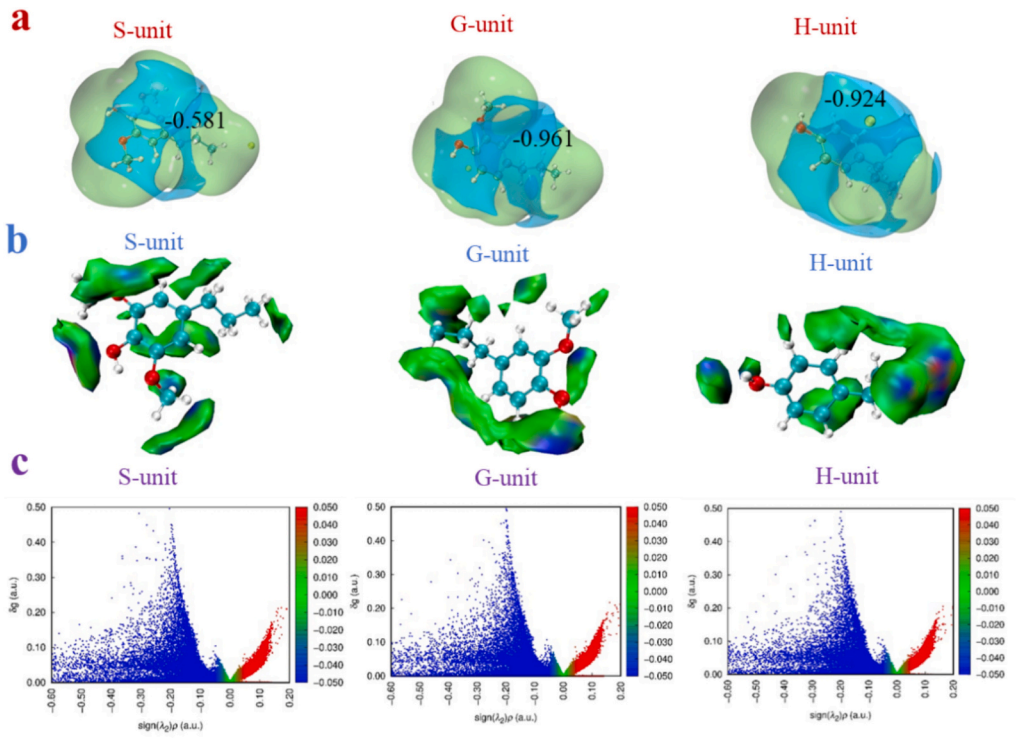


Fig. 5. The electrostatic potential pattern on VDW surface S-, G-, and H- unit of lignin models (a); The IGM iso-surface image of S-, G-, and H- units of lignin model with CTAC:LA (CTAC:LA=1:4) treatment system (b); The scatter plot of S-, G-, and H- units of lignin model with CTAC:LA (CTAC:LA=1:4) treatment system (c).

the blue area), and stronger hydrogen-bonds mean stronger reactivity between the two entities, confirming that the content of H-type lignin could decrease significantly after CTAC:LA treatment. In the scatter plot (Fig. 5c), the red, green, and blue regions are behalf of steric effects, van der Waals interactions, and hydrogen-bond interactions, respectively. There was a pronounced peak for the hydrogen-bond interaction between the three lignin structural units and the CTAC:LA treatment system. During the CTAC:LA treatment, the CTAC:LA solvent system might strongly interact with lignin through both van der Waals forces and hydrogen bonding and lead to substantial lignin removal, elucidating the mechanism of CTAC:LA treatment in molecular level.

3.5. Evaluation of reusability of CTAC:LA solvent system

The recovery and reuse capacity of DES is a crucial indicator for evaluating its pretreatment performance, which can enhance the economic feasibility of the biorefining process, and reduce the chemicals required for the reaction process and the waste generated [54]. The glucan content in untreated WS was 41.7 %. After the CTAC:LA treatment (160 °C, 60 min, CTAC:LA = 1:4), glucan content greatly increased to 66.2 % (Fig. 6a), indicating that CTAC:LA treatment effectively eliminated lignin (88.3 %) and xylan (89.0 %) in WS. The recycled CTAC:LA was then utilized for subsequent WS pretreatment. After the

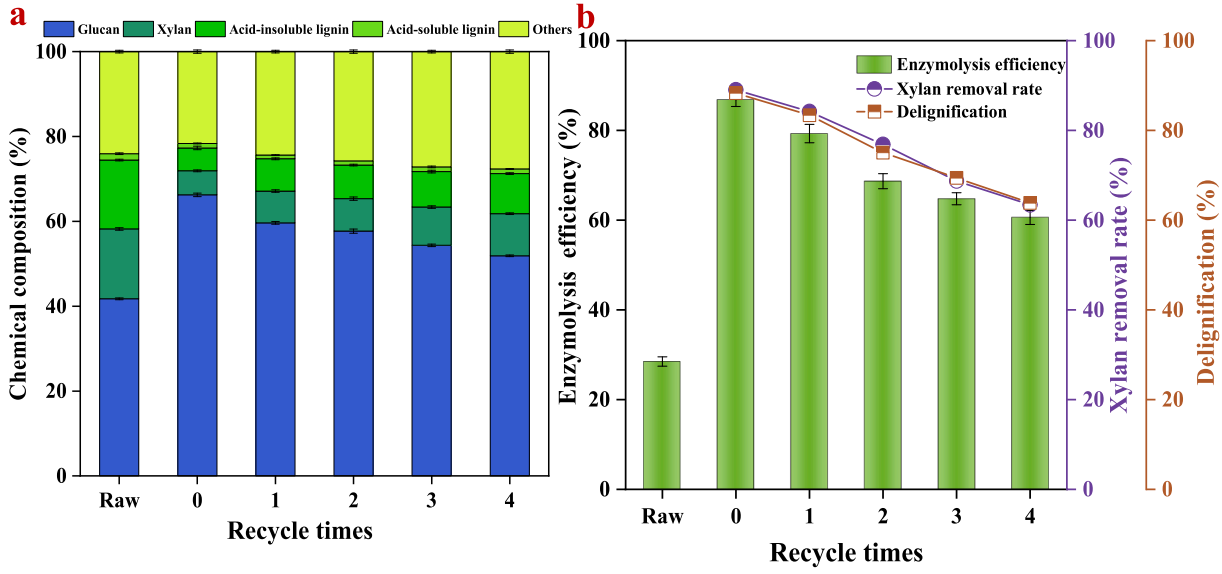


Fig. 6. The effect of recycle times of pretreatment system on chemical composition (a) and enzymolysis efficiency of WS (b) [pretreatment conditions: 160 °C, 60 min, CTAC:LA = 1:4].

first cycle, the lignin removal only decreased by 7.5 %, and the enzymolysis efficiency of WS after CTAC:LA treatment reached 79.3 %, indicating that recycled liquid still maintained a high pretreatment capability. In the cycle process, the accumulation of sugar or residues produced by substance fractionation affected the mass transfer process of WS pretreatment, and inevitably decreased the pretreatment effect with an increase in cycles [55]. Although there was a decrease in pretreatment efficiency after 4 cycles, the delignification removal still reached 63.8 %, and xylan removal reached 63.4 %. More importantly, the enzymolysis efficienccy of WS after 4 cycles of pretreatment was found to be 2.2 times that of original untreated WS (Fig. 6b), demonstrating superiority in terms of economic feasibility and sustainability for DES (CTAC:LA) solvent.

3.6. Properties of WS after CTAC:LA treatment

3.6.1. Analysis of accessibility

Previous studies have indicated that substrate properties, including the accessible area and crystallinity of the substrate, have important effects on the enzymolysis efficiency of lignocellulose [56–58]. The accessibility of raw materials was 306.7 mg/g, the low contactable degree of cellulose limited the conversion efficiency in the enzymatic hydrolysis process. After pretreatment with CTAC:LA, the accessibility of material increased from 378.9 mg/g (120 ℃, 60 min, CTAC:LA=1:2) to 625.1 mg/g (180 ℃, 60 min, CTAC:LA=1:2) (Fig. 7a). Under optimal pretreatment condition (160 °C, 60 min, CTAC:LA = 1:4), the accessibility was achieved 668.2 mg/g (Fig. 7c). The linear fitting between xylan removal [$Y_{(Accessibility)} = 5.81 \times X_{(Xylan\ removal)} + 118.57$, $R^2 = 0.92$] and delignification [$Y_{(Accessibility)} = 4.82 \times X_{(Delignification)} + 205.02$, $R^2 = 0.92$] with accessibility was explored in Fig. 7a. Lignin and hemicellulose act as physical barriers in lignocellulosic materials, preventing cellulase from approaching cellulose. CTAC:LA treatment removed large amounts of lignin and xylan from WS and greatly improved its accessibility, which in turn increased the chance of cellulose coming into contact with cellulases and being hydrolyzed, thereby increasing the biotransformation of cellulose to monosaccharides.

3.6.2. Analysis of lignin surface area

Lignin surface area is one of the important evaluation indexes for evaluating the pretreatment ability, as it renders a valuable insight for assessing structural alteration in lignocellulose [59]. As depicted in Fig. 7b, there existed a linear fitting about lignin surface area and

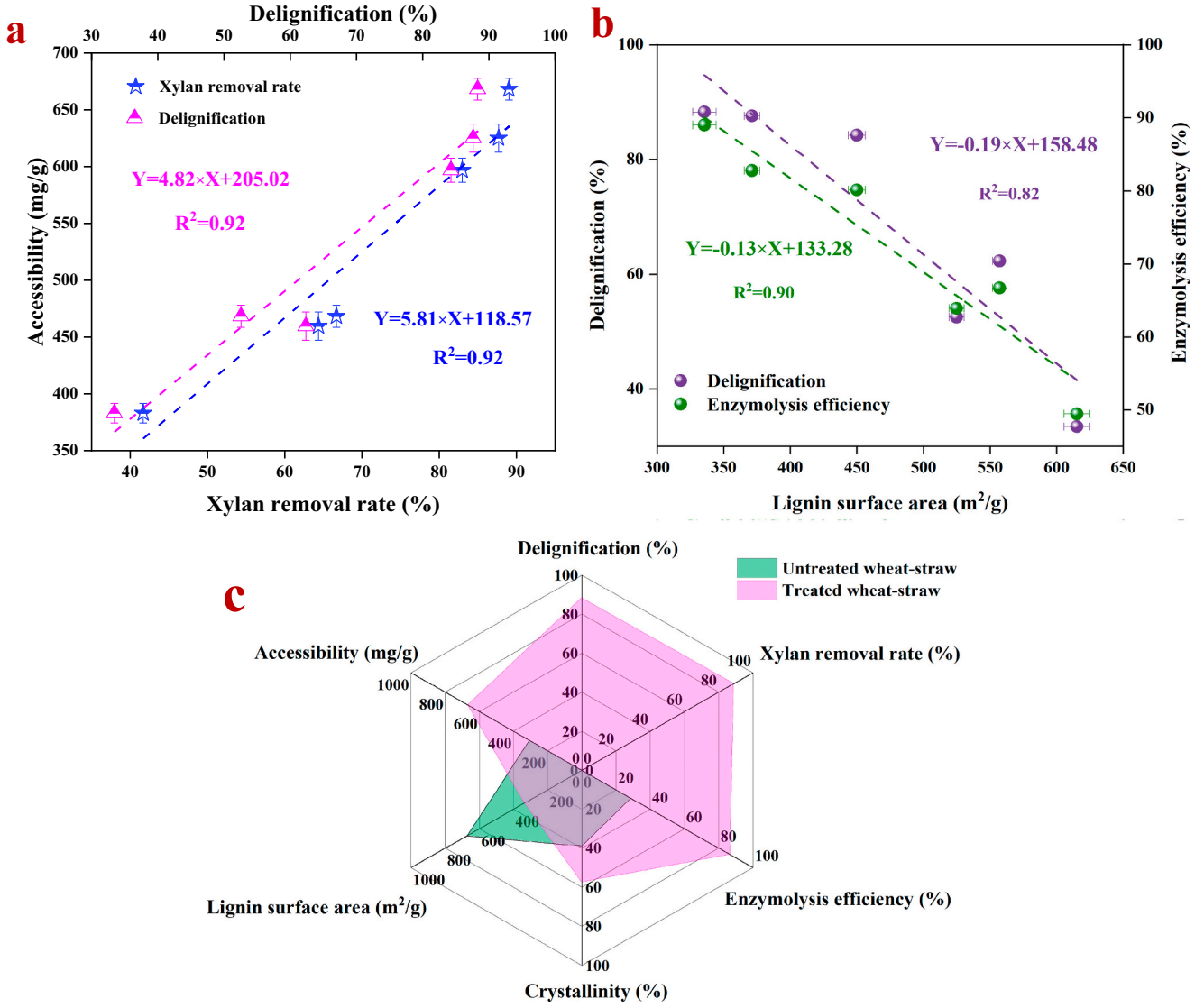


Fig. 7. The linear fittings about delignification and xylan removal with accessibility (a); The linear fitting about delignification, enzymolysis efficiency and lignin surface area (b). The radar image of the delignification, xylan removal, enzymolysis efficiency, accessibility and lignin surface area for both untreated WS (Raw) and WS after CTAC:LA treatment [pretreatment conditions: 160 ℃, 60 min, CTAC:LA=1:4] (c).

delignification [$Y_{(Delignification)} = -0.19 \times X_{(Lignin\ surface\ area)} + 158.48$, $R^2 = 0.82$]. The surface area of lignin in the raw material was 672.7 m^2/g. However, with the degradation and removal of lignin, the surface area of lignin decreased apparently, the lignin surface area of WS reduced to 335.4 m^2/g under the optimal pretreatment condition (160 °C, 60 min, CTAC:LA = 1:4) (Fig. 7c). The good linear relationship about lignin surface area and enzymolysis efficiency [$Y_{(Enzymolysis\ efficiency)} = -0.13 \times X_{(Lignin\ surface\ area)} + 133.28$, $R^2 = 0.90$] could be fitted (Fig. 7b). The complete cell structure was torn during treatment, large amounts of lignin and xylan were eliminated, the lignin surface area was greatly declined, and the polysaccharide structure of cellulose was rapidly exposed, resulting in an elevated enzymolysis efficiency.

3.6.3. Analysis of crystallinity

The crystallinity of lignocellulose is believed to play a crucial role in the conversion of lignocellulose into fermentable sugars [60]. The crystallinity of raw WS material reached 38.9 %, while the crystallinity was substantially elevated to 57.6 % after CTAC:LA treatment (160 °C, 60 min, CTAC:LA = 1:4) (Fig. S2, see supplementary file). The cellulose properties of the treated WS exhibited an increase in the crystallinity index compared with the untreated WS. CTAC:LA treatment eliminated large amounts of xylan and lignin in the WS and reduced their coating effect on cellulose, thus exposing crystalline cellulose and resulting in increased total crystallinity in the biomass [61]. This exposure of cellulose enhanced the contact area between the enzyme and cellulose, thereby accelerating the subsequent enzymolysis process. After CTAC:LA treatment (160 °C, 60 min, CTAC:LA = 1:4), the lignin removal reached 88.3 %, and xylan separation reached 89.0 %. The crystallinity and accessibility of lignocellulose were significantly improved, and the lignin surface area was reduced to 335.4 m^2/g (Fig. 7c). These positive changes greatly elevated the enzymolysis efficiency of WS after pretreatment.

3.6.4. SEM analysis of microstructure after CTAC:LA treatment

Scanning electron microscopy was utilized to observe the morphological and structural alterations of WS before and after CTAC:LA treatment (Fig. S3, see supplementary file). The surface morphology of the raw WS material was intact and dense, rendering protective layer for cellulose that makes it challenging for cellulase to access cellulose molecules. In contrast, the morphology of WS underwent significant alterations after CTAC:LA treatment. Following pretreatment, the material exhibited varying degrees of fragmentation, with a rough and porous surface along with some fragment-like structures. These alterations manifested that DES (CTAC:LA) pretreatment strategy

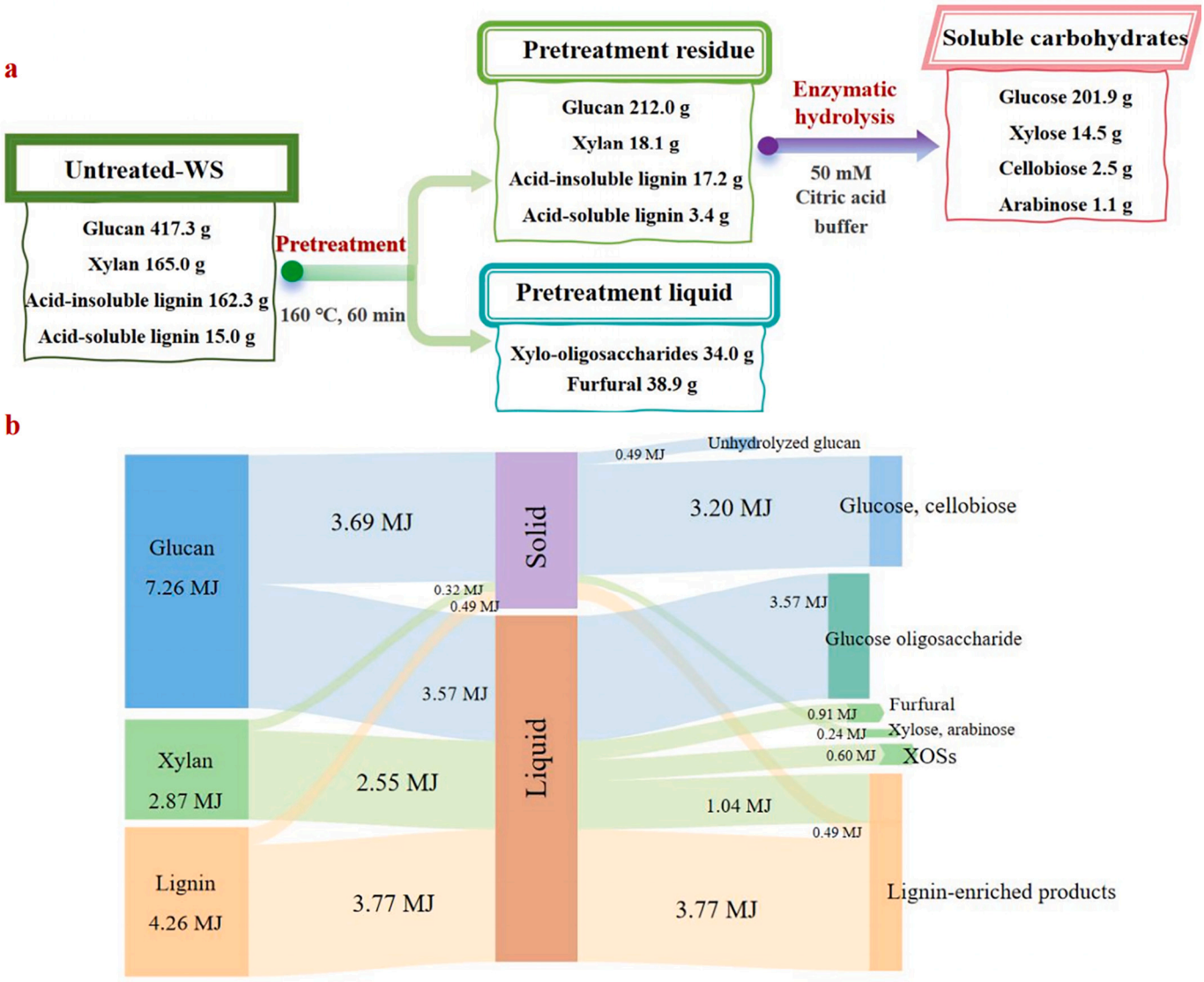


Fig. 8. Mass flow for CTAC:LA pretreatment of WS (a); Energy flow for the co-production of fermentable sugar, furfural and XOSs (b).

successfully disrupted the original physical structure of the WS, enhancing the number of reaction sites for cellulase.

3.7. Mass flow and energy flow

Lignocellulose can be effectually transformed into fermentable sugars and other bio-based products through efficient pretreatment and enzymolysis [62,63]. In this research, 1000 g WS composed of 417.3 g glucan, 165.0 g xylan, 162.3 g acid-insoluble lignin, and 15.0 g acid-soluble lignin was used as raw material for biorefining. After CTAC:LA fractionation, the cellulose-rich solid residue consisting of 212.0 g glucan, 18.1 g xylan, 17.2 g acid-insoluble lignin, and 3.4 g acid-soluble lignin was recovered. The removal of xylan was 89.0 % and the lignin removal reached 88.3 %. In the pretreatment liquid, 38.9 g furfural and 34.0 g xylo-oligosaccharides were produced. After enzymolysis of the cellulose-rich solid residue, 201.9 g glucose, 2.5 g cellobiose, 14.5 g xylose, 1.1 g arabinose were released (Fig. 8a). These sugars are important bio-based products that can be used in subsequent biotransformation processes. To explore the potential applications of this biorefining pathway, the overall energy flow was further evaluated (Fig. 8b). The energy flow image manifested that the high calorific value (HHV) of WS consists of glucan (7.26 MJ/kg WS), xylan (2.87 MJ/kg WS), and lignin (4.26 MJ/kg WS). Throughout the entire biorefining process of WS (1000 g), the energy output of furfural, XOSs, and lignin-enriched products reached 0.91, 0.60, and 5.30 MJ respectively. The majority of carbohydrate energy has been successfully converted into value-added forms, demonstrating the potential industrial application. Despite the remarkable progress in the biomass utilization and conversion process, the generation of lignin-enriched products after pretreatment was still an unavoidable problem. Lignin has the potential for many applications [64]. Li et al. found that the lignin regenerated from the acid DES (ChCl:oxalic acid) pretreatment solution had good thermal stability and antioxidant activity [65]. In addition, Liu et al. showed that biobased plastics prepared with recycled lignin from the pretreatment solution have better mechanical and tensile strength [66]. Therefore, these lignin-rich products generated in the CTAC:LA pretreatment system provide unlimited possibilities for further processing and upgrading into high value-added products (such as antioxidants, heat stabilizers, and plastics).

3.8. Potential CTAC:LA pretreatment mechanism

Effective pretreatment of WS was realized through the action of DES (CTAC:LA), and its potential pretreatment mechanism was illustrated in Fig. 9. As a carboxylic acid with high hydrogen bond donor capacity, lactic acid might provide large numbers of protons for efficient elimination of hemicellulose and lignin [12]. CTAC, a surfactant with hydrophobic long alkyl chain structure, not only affected the hydrophobic lignin structure but also could prevent the decomposed lignin molecules from recondensing into pseudo-lignin and being deposited on the biomass surface [53]. Under the action of CTAC:LA treatment system, a large number of lignin and xylan were eliminated from the WS structure, which substantially elevated the accessibility and crystallinity of WS after treatment, and reduced the lignin surface area, thus advancing the subsequent enzymolysis efficiency (Fig. 9). Through quantum chemistry calculation and molecular dynamics simulation, it was found that there was strong and weak interaction (mainly van der Waals forces and hydrogen bond interaction) between lignin model and CTAC:LA treatment system. These interactions lead to the weakened connections between lignin, hemicellulose and cellulose, and the S-units, H-units, C—O (β-O-4) covalent and C—C (β-5, β-β) bonds in the lignin were effectually cleaved by CTAC:LA, promoting the removal of lignin and thereby enhancing the enzymolysis of carbohydrates.

The key to making the most of lignocellulose and achieving an environmentally sustainable future lies in effective pretreatment strategies [67–69]. In this research, CTAC containing Cl- was used as hydrogen bond receptor to synthesize a new type of cationic surfactant-based DES (CTAC:LA). This system could efficiently remove lignin and xylan from WS, significantly improve the accessibility and crystallinity of WS after pretreatment and make the substrate after pretreatment show excellent enzymatic hydrolysis efficiency (86.8 %). This provided favorable conditions for the conversion of biomass into high-value chemicals. In addition, various characterizations combined with molecular simulation revealed the potential pretreatment mechanism of cationic surfactant-based DES, which provided theoretical support for the construction of the lignocellulosic pretreatment process. Liu et al. proposed that the recycling of DES has always been controversial, because its recycling process is complex, and the efficiency of DES after recycling decreases with the increase of the recycling cycle [70]. Anti-solvent addition, membrane separation, and electrodialysis are common means of separating and recovering DES after fractionation of biomass, however, they do not seem to be a good option in terms of cost and ease of use [71]. Simple recycling methods such as filtration and centrifugation cannot separate the lignin dissolved in DES, which significantly affects the pretreatment performance of DES after recovery. Therefore, how to remove impurities while maintaining the original performance of DES is a problem that needs to be solved at present.

4. Conclusion

In this study, the novel cationic surfactant-based DES system was developed by cetyl trimethyl ammonium chloride and lactic acid. The CTAC:LA system effectively removed significant amounts of lignin (88.3 %) and xylan (89.0 %) from WS under the optimal pretreatment conditions [160 °C, 60 min, CTAC:LA = 1:4]. After pretreatment, the lignin surface area of WS decreased significantly, the accessibility and

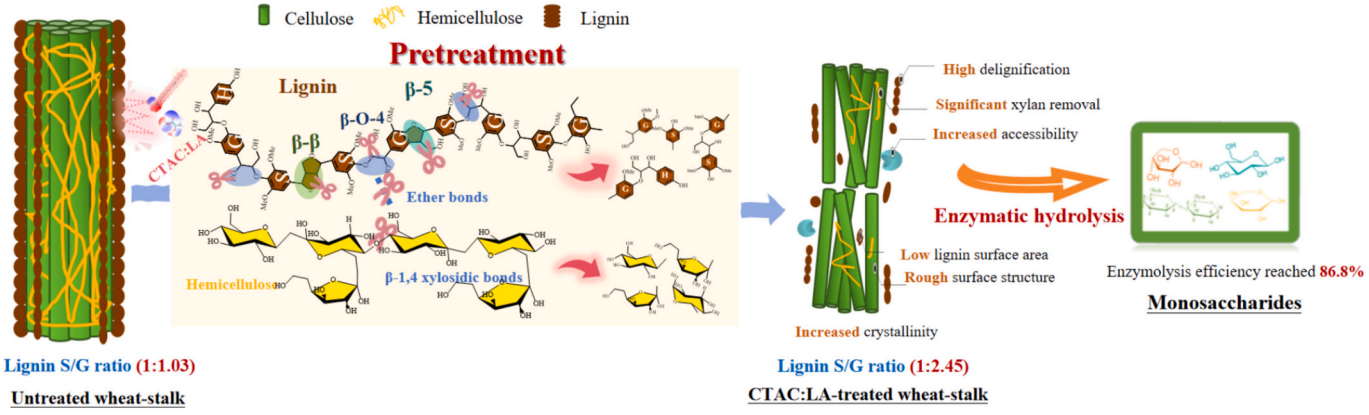


Fig. 9. Potential mechanism of CTAC:LA pretreatment to enhance enzymolysis efficiency of wheat-stalk.

crystallinity increased notably, and the surface structure became rough and porous. An inverse "hook-shape" fitting between the content of XOSs and severity factors was observed. There was positive fitting between the severity factors and furfural production. In addition, the relationships of accessibility, delignification, xylan removal and enzymolysis efficiency were explored. Molecular dynamics simulations revealed strong hydrogen bond interaction and van der Waals interaction between CTAC:LA system and lignin. CTAC:LA system demonstrated efficient cleavage of C—O covalent bonds (β-O-4) and C—C bonds (β-5, β-β) in lignin, as well as degradation of S- and H-units in WS lignin through interaction, which supplemented the potential mechanism of DES pretreatment. Consequently, these positive structural changes made the cellulose-rich substrate pretreated with CTAC:LA have good enzymolysis efficiency (86.8 %). The treatment with CTAC:LA realized efficient fractionation of lignocellulose, and the abundant lignin fragments open up new opportunities for lignin appreciation, which contributes to the more comprehensive utilization of lignocellulose resources.

CRediT authorship contribution statement

Ying Chen: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization, Software. **Zhengyu Tang:** Investigation, Software. **Wei Tang:** Methodology, Investigation. **Cuiluan Ma:** Methodology, Data curation. **Yu-Cai He:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.141460.

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